

Enrico VERALDI



PROFILE

I am a **PhD student in Astrophysics and Cosmology at SISSA (Trieste)** and a **master's student in High Performance Computing (MHPC) at ICTP (Trieste)**. My main research interests regard the characterization of the physical properties of the interstellar medium in high-redshift galaxies and the creation of associated models.

CONTACT DETAILS

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15/09/199, Bologna (Italy), He-Him

PERSONAL INFORMATION

Citizenship: **Italian**
Family: **Single without children**
Languages: **English** (B2),
Italian (native)

TECH STACK

- Python, Fortran, C/C++, CUDA, Bash, R
- Numpyh, Pandas, Matplotlib, Scikit-learn, PyTorch
- LaTeX, Docker, Anaconda
- Hugging Face, Ollama
- CLOUDY, NRAO CASA, ds9
- MPI, OpenMP, OpenACC
- Linux Systems, Slurm, Base System Adm.

PRESENT POSITIONS

- PhD. Student**
SISSA, Trieste (TS). 2024.10–pres.
◊ PhD in Astrophysics and Cosmology (APC sector)
- Master's Student**
MHPC IFTP-SISSA, Trieste (TS). 2025.09–pres.
◊ 2nd level master's in high performance computing

EDUCATION

- Master in Astrophysics and Cosmology**
University of Bologna (UNIBO). 2021–2023
◊ Thesis title: *The ISM properties of ALPINE galaxies as probed by ALMA and JWST: a comprehensive line emission modelling.*
◊ Supervisors: Prof. Francesca Pozzi, Dr. Livia Vallini, Dr. Federico Esposito.
- Bachelor in Physics**
University of Bologna (UNIBO). 2018–2021
◊ Thesis title: *Study of a spiral galaxy in the Local Universe.*
◊ Supervisors: Prof. Francesca Pozzi, Dr. Viviana Casasola.
- High school graduation**
Scientific High School Niccolò Copernico (Bologna). 2018–2021
◊ Scientific Applied Sciences

FIRST-AUTHOR PUBLICATIONS

- The ALPINE-ALMA [CII] Survey: Modelling ALMA and JWST lines to constrain the interstellar medium of $z \sim 5$ galaxies** 2025.01
◊ Veraldi et al. 2025, DOI: [10.1051/0004-6361/202451111](https://doi.org/10.1051/0004-6361/202451111)

CO-AUTHOR PUBLICATIONS

- The ALPINE-CRISTAL-JWST Survey: The Fast Metal Enrichment of Massive Galaxies at $z \sim 5$** 2026.06
◊ Faisst et al. 2026, DOI: [10.3847/1538-4357/ae6645](https://doi.org/10.3847/1538-4357/ae6645)
- The ALPINE-CRISTAL-JWST Survey: Chemical Abundance Comparison between the ISM and CGM of Main-sequence Galaxies at $z = 4 - 6$** 2026.04
◊ Wang et al. 2026, DOI: [10.3847/1538-4357/ae4b38](https://doi.org/10.3847/1538-4357/ae4b38)
- The ALPINE-CRISTAL-JWST Survey: JWST/IFU Optical Observations for 18 Main-sequence Galaxies at $z = 4 - 6$** 2026.01
◊ Faisst et al. 2026, DOI: [10.3847/1538-4365/ae0928](https://doi.org/10.3847/1538-4365/ae0928)
- The ALPINE-CRISTAL-JWST survey: revealing less massive black holes in high-redshift galaxies** 2025.11
◊ Ren et al. 2025, DOI: [10.1093/mnras/staf1709](https://doi.org/10.1093/mnras/staf1709)
- The ALPINE-CRISTAL-JWST survey: Spatially resolved star formation relations at $z \sim 5$** 2025.10
◊ Accard et al. 2025, DOI: [10.1051/0004-6361/202556140](https://doi.org/10.1051/0004-6361/202556140)
- Spatially resolved [CII]-gas conversion factor in early galaxies** 2025.08
◊ Vallini et al. 2025, DOI: [10.1051/0004-6361/202555179](https://doi.org/10.1051/0004-6361/202555179)

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REFEREES

@ Prof. Andrea Lapi

@ Dr. Tommaso Ronconi

@ Dr. Livia Vallini

@ Prof. Francesca Pozzi

POSTERS/TALKS/SEMINARS

Seminar "A Panchromatic View of the Early ISM: Modelling UV, Optical, and FIR Lines at $z \sim 5$ connecting ALMA and JWST" **2026.05**

◇ Seminar: "Astrophysics talk", INAF-OAS Bologna (Italy) ([website](#))

Talk "The ALPINE-ALMA [CII] Survey: Modelling ALMA and JWST lines to characterise the interstellar medium of high-redshift galaxies" **2025.12**

◇ Conference: "The Baryon Cycle from Reionization to the Cosmic Noon", Puerto Varas (Cile) ([website](#))

Poster-Flash talk "The ALPINE-ALMA [CII] Survey: Modelling ALMA and JWST lines to characterise the interstellar medium of high-redshift galaxies" **2025.06**

◇ Conference: "Sixth Workshop on Millimetre Astronomy in Italy", Bologna ([ARC](#))

Poster-Flash talk "The ISM properties of ALPINE galaxies as probed by ALMA and JWST" **2024.09**

◇ Conference: "Views on the multi-phase interstellar medium in galaxies", Bologna ([poster](#))

SCHOOLS/WORKSHOPS

Summer School "Statistics for Astronomers" **2026.06**

◇ Center for Astrostatistics at The Pennsylvania State University, PA 16802 (USA)

Workshop "Probing Galaxy Evolution at Cosmic Noon with Lensed DSFGs: A Multi-Wavelength Approach from Far-infrared to Radio" **2026.04**

◇ IFPU, Trieste (Italy) ([website](#))

SCIENTIFIC COLLABORATIONS

Member GoTha Group, SISSA **2025-pres.**

Member ALPINE-CRISTAL Collaboration **2024-pres.**

PROPOSALS

PROJECTS/FUNDING/GRANTS